Samuel Granados

BLVD. Primavera 3229, Monterrey, N.L. 64830 | +52 642 145 1046 | linkedin.com/in/samuel-granados | a00228152@tec.mx

Education

Instituto Tecnológico y de Estudios Superiores de Monterrey — **Monterrey, N.L.**
B.S. in Electronics Engineering — *Expected Graduation: December 2027*

- GPA 3.8/4.0 | Full Academic Leadership Scholarship (100%)
- Specialization: Hardware design & Power electronics

Hong Kong University of Science and Technology (HKUST) — **Hong Kong**
Exchange Program — *Expected: Fall 2026*

Work Experience

Carrier — **Santa Catarina, N.L.**
Electronics Validation Automation Intern | Electrical CoE — *June 2025 – March 2026*

- **Multiplexer:** Designed an RS-485 protocol multiplexer utilizing an ATmega328 MCU and DG409 analog multiplexer on a custom Altium PCB; developed a C# graphical interface to evaluate real-time data from boards sharing the same address.
- **Root cause analysis:** Conducted Root Cause Analysis (RCA) on PCB failures, proposing component or design modifications and replicating legacy failures that remained unresolved for over a year.
- **Component validation test:** Performed component validation testing for PCB production lines with environmental chambers using HIL test rigs interfaces for thermal shocks and monitoring with C# interface and deploying it in Power BI.

Independent Test & Validation Intern | Software & Controls CoE — *March 2026 – Present*

- **Small Sim:** Designed in Altium a shield for an STM32H563ZI microcontroller leveraging FreeRTOS; replicated residential AC external sensor inputs to validate embedded firmware using 24VAC and PWM signal generation.

Co-curricular Activities & Achievements

RoBorregos — **Monterrey, N.L.**
Director of Finance — *August 2025 – Present*

- Led a 10-member team in fundraising and budget planning, securing $400K+ MXN in corporate sponsorships and managing a $700K+ MXN total budget for national and international competitions, including RoboCup 2026 and LARC 2025.

Electronics PM – @HOME League — *June 2025 – Present*
Mexican Tournament of Robotics and RoboCup 2026

- Led the electronic design for a 1200W power distribution and battery charging system; modified and integrated a custom BMS and distribution board within an 8-engineer team, achieving **1st Place Nationally** (April 2026).

Electronics Member | Rescue Maze League — *November 2024 – May 2025*
Mexican Tournament of Robotics

- Designed electronics under stringent size and energy density constraints, resulting in a system architecture that directly contributed to achieving **2nd Place Nationally** at the Mexican Tournament of Robotics (May 2025).

PLEI Asia Mission — **International**
Selected Mission Member — *June 2026 – August 2026*

- Selected as the first fully sponsored student to join an international business and leadership mission (summer 2026), conducting market research across East and Southeast Asia (China, Taiwan, Indonesia, Thailand, Malaysia, and Vietnam).

EcoShell Tec Racing — **Monterrey, N.L.**
Electronics Team Member — *August 2024 – June 2025*

- Developed vehicle telemetry systems using UART and I2C protocols with ESP32 microcontrollers, enabling real-time vehicle status acquisition and data analysis in MATLAB.

Industry-Sponsored Engineering Projects in School

- **OPmobility:** Designed a protection and control system for automotive lighting using a PIC18 microcontroller, including fault detection and system validation using mirror currents.
- **Bendix:** Designed and simulated a custom inductor using COMSOL Multiphysics, optimizing magnetic performance and validating design constraints in collaboration with industry engineers.
- **Acuity Brands:** Designed an analog signal conditioning circuit using operational amplifiers to control a motor-driven window blind, focusing on gain, filtering, stability, and noise reduction without embedded firmware.
- **DIRAM:** Developed and tuned a PID-based position control system, analyzing system response to meet industrial performance criteria.

Skills and Interests

Programming & Control: C++, C#, C, Python, MATLAB, Simulink.

Embedded Systems: Microcontrollers (ESP32, PIC18, TI F28379D, STM32), Data Acquisition (DAQ) Systems, Verilog.

Communication Protocols: CAN, RS-485, I2C, UART, TCP/IP, SCPI.

Design: Hardware design (Altium, KiCad, OrCad), Simulation (LTspice, Multisim).

Electronics: Analog IC Design Fundamentals, ADC/DAC Principles, Semiconductor Devices, Optoelectronics, Amplifiers, Power Electronics, Telecommunications, Signal processing.

Languages: English (B2 - 6.5 IELTS), Spanish (Native).

Certifications: Google Data Analytics (2025). Lean Six Sigma Green Belt (2024). MIT NanoLab - Intro to Micro and Nano Production (2024).